

HACK: Supplementary Material

Additional accuracy results for “HACK: Homomorphic Acceleration via Compression of the Key-Value Cache for Disaggregated LLM Inference” (ACM SIGOPS ATC 2026)

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This document supplements the paper “HACK: Homomorphic Acceleration via Compression of the Key-Value Cache for Disaggregated LLM Inference” (ACM SIGOPS ATC 2026, <https://doi.org/10.1145/3828523.3857108>) with the per-dataset accuracy tables summarized in its §6.3. The code of HACK is at <https://github.com/Zeyu-ZEYU/HACK> (archived at <https://doi.org/10.5281/zenodo.22982679>). Table numbers below are local to this document.

1 Additional Accuracy Results

Tables 1–6 report per-dataset accuracy across all evaluated models; Table 7 reports additional metrics (perplexity, ROUGE-L, and pass@1/pass@10) for Qwen3 32B.

Table 1. Accuracy (IMDb).

	Mis	Phi	Qwen	Yi	Lla	Fal
Baseline	84.81%	87.84%	93.15%	93.87%	95.73%	85.63%
HACK ($\Pi=32$)	83.79%	87.08%	93.02%	92.82%	94.73%	84.96%
HACK ($\Pi=64$)	83.46%	86.64%	92.48%	92.31%	94.17%	84.72%
CacheGen	83.04%	86.15%	91.19%	91.79%	93.85%	83.94%
KVQuant	82.97%	86.05%	91.10%	91.54%	93.89%	84.12%
HACK ($\Pi=128$)	82.71%	86.07%	90.75%	91.35%	93.76%	84.03%

Table 2. Accuracy (arXiv).

	Mis	Phi	Qwen	Yi	Lla	Fal
Baseline	79.40%	86.35%	88.50%	87.75%	83.79%	79.42%
HACK ($\Pi=32$)	78.53%	85.51%	87.73%	86.69%	82.63%	78.75%
HACK ($\Pi=64$)	78.26%	85.08%	87.25%	86.39%	82.31%	78.28%
CacheGen	77.95%	84.02%	85.74%	85.14%	81.85%	77.74%
KVQuant	77.64%	84.11%	85.50%	85.08%	81.76%	77.72%
HACK ($\Pi=128$)	77.72%	84.49%	86.92%	86.15%	81.97%	77.94%

Table 3. Accuracy (Cocktail).

	Mis	Phi	Qwen	Yi	Lla
Baseline	75.18%	83.92%	89.25%	85.25%	86.39%
HACK ($\Pi=32$)	74.63%	83.05%	88.47%	84.39%	85.54%
HACK ($\Pi=64$)	74.42%	82.66%	88.06%	84.04%	84.87%
CacheGen	73.74%	82.24%	87.61%	83.51%	84.71%
KVQuant	73.72%	82.26%	87.58%	83.42%	84.68%
HACK ($\Pi=128$)	73.81%	82.19%	87.44%	82.86%	84.65%

Table 4. Accuracy (HumanEval).

	Mis	Phi	Qwen	Yi	Lla	Fal
Baseline	89.37%	91.62%	92.17%	90.79%	92.45%	85.21%
HACK ($\Pi=32$)	88.68%	90.50%	91.23%	90.02%	91.28%	84.26%
HACK ($\Pi=64$)	88.19%	90.16%	90.85%	89.63%	91.04%	83.82%
CacheGen	87.73%	89.75%	90.36%	88.94%	90.49%	83.47%
KVQuant	87.64%	89.65%	90.27%	88.93%	90.43%	83.29%
HACK ($\Pi=128$)	87.60%	89.54%	90.08%	88.91%	89.77%	83.19%

Table 5. Accuracy (GSM8K).

	Mis	Phi	Qwen	Yi	Lla	Fal
Baseline	96.1%	94.3%	95.9%	96.2%	95.6%	83.7%
HACK ($\Pi=32$)	95.4%	93.6%	95.0%	95.2%	94.5%	82.8%
HACK ($\Pi=64$)	95.2%	93.2%	94.6%	94.7%	94.1%	82.3%
CacheGen	94.6%	92.8%	94.2%	94.4%	93.7%	81.7%
KVQuant	94.5%	92.7%	94.1%	94.4%	93.5%	81.6%
HACK ($\Pi=128$)	94.2%	92.3%	93.6%	93.8%	93.1%	81.2%

Table 6. Accuracy (RULER-128K).

	Mis	Phi	Qwen	Yi	Lla
Baseline	48.1%	55.7%	85.6%	77.3%	66.6%
HACK ($\Pi=32$)	47.7%	55.2%	85.1%	76.9%	66.2%
HACK ($\Pi=64$)	46.9%	54.7%	84.5%	76.0%	65.4%
CacheGen	46.0%	53.9%	83.7%	75.4%	64.8%
KVQuant	46.2%	53.6%	83.2%	75.5%	64.3%
HACK ($\Pi=128$)	45.1%	52.8%	82.4%	74.2%	63.6%

Additional metrics. Beyond exact match, ROUGE-1, and edit similarity, Table 7 reports perplexity (PPL, WikiText-2 [1]), ROUGE-L (arXiv summarization), and pass@1/pass@10 (HumanEval) for Qwen3 32B. HACK ($\Pi=32$) and HACK ($\Pi=64$) track the BF16 baseline most closely and beat CacheGen and KVQuant on all four metrics, with the smaller partition slightly ahead, consistent with the exact-match results above.

Table 7. Additional accuracy metrics for Qwen3 32B.

	WikiText-2	arXiv	HumanEval	
Method	PPL ↓	ROUGE-L	pass@1	pass@10
Baseline	7.6	81.0%	87.0%	93.0%
CacheGen	8.2	78.0%	83.5%	90.5%
KVQuant	8.3	77.8%	83.0%	90.2%
HACK ($\Pi=32$)	7.8	80.2%	86.0%	92.4%
HACK ($\Pi=64$)	7.9	79.6%	85.3%	91.9%

References

- [1] Stephen Merity, Caiming Xiong, James Bradbury, and Richard Socher. 2017. Pointer Sentinel Mixture Models. In *International Conference on Learning Representations (ICLR)*. <https://arxiv.org/abs/1609.07843>